ME 4805 – Power Plant Engineering

CEP Project

Base paper:

"Investigation of a novel solar-assisted multigeneration system comprising water desalination systems, an absorption refrigeration system, a single-effect absorption heat transformer and organic Rankine cycles"

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1 Executive Summary

This report presents an independent validation of the thermodynamic model published by Salehi and Javanfam [1], which describes a solar-driven multigeneration system that produces electricity, refrigeration, and desalinated water from a single set of solar collectors. The system integrates an absorption refrigeration (AR) unit, a single-effect absorption heat transformer (AHT), two organic Rankine cycles (ORC1 and ORC2), and two humidification–dehumidification (HDH) desalination loops, together forming 61 numbered state points.

To validate this model, the system was rebuilt independently in Engineering Equation Solver (EES) as four subsystem codes – the water loop and ORC2 (state points 1–9), the AR system together with the open- and closed-loop HDH units (state points 10–38), the AHT (state points 39–50), and the evaporative-desalination separator together with ORC1 (state points 51–61) – which were then combined into a single integrated model spanning all 61 state points. Each subsystem was solved using the same mass, species, and energy balance equations, and the same operating conditions (temperatures, pressures, effectiveness values, and solar heat input), as reported in the base paper, with standard seawater property correlations [3] used wherever the paper itself did not specify a correlation.

The calculated state-point properties – mass flow rate, temperature, pressure, enthalpy, entropy, vapor quality, relative humidity, humidity ratio, salinity, and LiBr concentration – were then compared directly against the base paper's published Table 4, state point by state point. Because the base paper does not provide enough information to reconstruct most of its parametric studies, the validation focused mainly on this state-point comparison. A smaller set of parametric checks was also run on the individual AR and AHT subsystems (COP versus generator, condenser, and absorber temperature) to confirm that they reproduce the trends expected from the wider literature, cross-checked against the single-effect LiBr absorption refrigeration study of Modi et al. [2].

Section 2 shows the schematic of the complete system. Section 3 presents the resulting state-point comparison, split into two tables for readability, with a short comment on the level of agreement after each. Section 4 presents the validated parametric trend graphs. Section 5 summarizes the specific inconsistencies found in the base paper during this process.

2 Schematic Diagram

Fig. 1 shows the schematic layout of the complete solar-assisted multigeneration system, including all four subsystems.

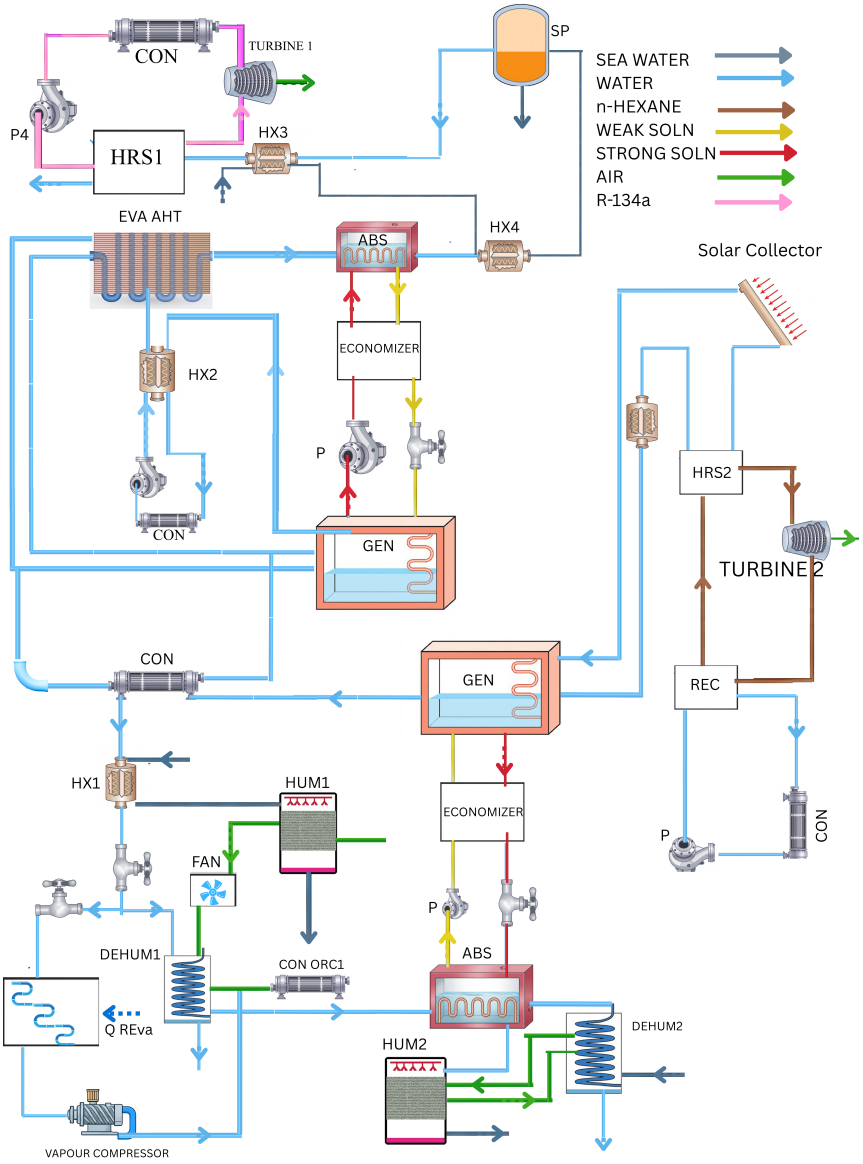


Figure 1: Schematic diagram of the solar-assisted multigeneration system, showing the absorption refrigeration unit, the single-effect absorption heat transformer, the two organic Rankine cycles, and the two humidification–dehumidification desalination loops.

3 Comparison Table

The state-point comparison follows the base paper's Table 4 [1], listing the calculated ("Calc") value from the present EES model next to the corresponding published ("Paper") value for all 61 state points. Because the full table is too wide for a single page, it is split into two parts: Table 1 covers mass flow rate, temperature, pressure, enthalpy, and entropy; Table 2 covers relative humidity, humidity ratio, salinity, and LiBr concentration.

Table 1: State-point comparison — mass flow rate, temperature, pressure, enthalpy, and entropy (calculated vs. base-paper Table 4 values).

State	$\dot{m}$ (kg/s)		T (°C)		P (kPa)		h (kJ/kg)		s (kJ/kg K)	
	Calc	Paper	Calc	Paper	Calc	Paper	Calc	Paper	Calc	Paper
1	2.332	2.329	150	150	476.2	475.7	632.2	632.3	1.842	1.842
2	2.332	2.329	135	135	476.2	475.7	567.9	567.9	1.687	1.687
3	2.332	2.329	127.5	127.5	476.2	475.7	535.9	535.9	1.608	1.607
4	0.1647	0.165	130.2	130.2	488.4	488.8	541.6	541.7	1.439	1.439
5	0.1647	0.165	80.68	80.7	25	25	460.7	460.8	1.497	1.497
6	0.1647	0.165	37.8	37.8	25	25	382.7	382.7	1.263	1.263
7	0.1647	0.165	30	30	25	25	7.1	7.1	0.024	0.024
8	0.1647	0.165	30.21	30.2	488.4	488.8	8	8	.02434	0.025
9	0.1647	0.165	63.5	63.5	488.4	488.8	86.1	86.1	0.2687	0.268
10	0.0434	0.053	130	130	57.87	57.81	2741	2740	7.778	7.777
11	0.0434	0.053	85	85	57.87	57.81	356	355.9	1.135	1.134
12	0.0434	0.053	65.95	66.1	57.87	57.81	276.1	276.7	0.905	0.907
13	0.0434	0.053	20	20	2.34	2.34	276.1	276.7	0.952	0.954
14	0.0262	0.032	15	15	1.71	1.71	276.1	276.7	0.964	0.966
15	0.0262	0.032	15	15	1.71	1.71	2528	2528	8.78	8.779
16	0.0262	0.032	44.06	40.9	2.34	2.34	2583	2577	8.815	8.795
17	0.0172	0.021	20	20	2.34	2.34	2537	2537	8.666	8.665
18	0.0123	0.015	23.65	21	101.3	101.3	99.2	88.1	0.348	0.31
19	0.0434	0.053	34.51	32.6	2.34	2.34	2565	2561	8.757	8.744
20	0.9015	1.137	50	50	2.34	2.34	113.2	112.3	–	0.316
21	0.9015	1.137	50	50.1	57.87	57.81	113.2	113.1	–	0.315
22	0.9015	1.137	108.7	122.7	57.87	57.81	235.5	267.7	–	0.747
23	0.8581	1.085	130	130	57.87	57.81	283.6	286.2	–	0.752
24	0.8581	1.085	66.02	51	57.87	57.81	155.1	123.2	–	0.304
25	0.8581	1.085	66.02	55.2	2.34	2.34	155.1	123.2	–	0.33
26	3.451	3.451	25	25	101.3	101.3	99.8	99.7	0.35	0.35
27	3.451	3.451	34.99	37.8	101.3	101.3	139.7	151.1	0.482	0.519
28	3.451	3.451	45.29	48	101.3	101.3	180.9	191.8	0.613	0.647
29	1.419	1.419	28.99	30.5	101.3	101.3	94.6	102.4	5.938	5.963
30	3.404	3.395	35.42	35.3	101.3	101.3	141.3	140.7	0.487	0.485
31	1.466	1.474	44.4	46.8	101.3	101.3	198.2	223.4	6.273	6.352
32	0.0468	0.055	28.99	30.5	101.3	101.3	121.5	127.8	0.423	0.443
33	0.018	0.018	25	25	101.3	101.3	99.8	99.8	0.35	0.35
34	0.018	0.018	73.04	84	101.3	101.3	292.5	336.6	0.949	1.075
35	0.014	0.014	45.95	47.9	101.3	101.3	181.1	188.8	0.611	0.634
36	0.904	0.904	41	41	101.3	101.3	113.3	113.4	5.999	5.999
37	0.908	0.908	32.98	33	101.3	101.3	116.4	116.6	6.01	6.009
38	0.8956	0.893	23.65	21	101.3	101.3	70.8	60.9	5.858	5.824
39	0.021	0.026	80	80	4.25	4.25	2650	2560	8.741	8.74
40	0.021	0.026	39.94	30.1	4.25	4.25	2574	2556	8.513	8.452
41	0.021	0.026	30	30	4.25	4.25	125.7	125.7	0.437	0.437
42	0.021	0.026	30	30	47.42	47.37	125.8	125.7	0.437	0.437

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Table 1 continued from previous page

State	$\dot{m}$ (kg/s)		T (°C)		P (kPa)		h (kJ/kg)		s (kJ/kg K)	
	Calc	Paper	Calc	Paper	Calc	Paper	Calc	Paper	Calc	Paper
43	0.021	0.026	48.13	52.5	47.42	47.37	201.6	220	0.68	0.737
44	0.021	0.026	80	80	47.42	47.37	2643	2643	7.611	7.611
45	0.0811	0.095	103	103	47.42	47.37	223.4	229	0.726	0.748
46	0.0811	0.095	92.12	91.1	4.25	4.25	198.4	200.7	0.661	0.672
47	0.0811	0.095	50.94	50.3	4.25	4.25	198.4	200.7	0.684	0.688
48	0.0601	0.069	80	80	4.25	4.25	208.8	210.2	0.429	0.435
49	0.0601	0.069	80	80.1	47.42	47.37	208.8	210.5	0.429	0.436
50	0.0601	0.069	98.4	100.8	47.42	47.37	242.5	248.9	0.517	0.541
51	0.0341	0.042	25	25	101.3	101.3	99.8	100.33	0.35	0.357
52	0.0341	0.042	86.42	90.1	101.3	101.3	346.4	379.3	1.102	1.19
53	0.0341	0.042	101.7	101.8	101.3	101.3	1873	1872.4	5.186	5.166
54	0.0222	0.027	101.7	101.8	101.3	101.3	2679	2678.1	7.364	7.331
55	0.0119	0.015	101.7	101.8	101.3	101.3	375.1	375.3	1.14	1.141
56	0.0222	0.027	99.97	100	101.3	101.3	2300	2251.3	6.347	6.216
57	0.0222	0.027	37.06	37.2	101.3	101.3	155.3	155.9	0.533	0.535
58	0.2703	0.328	32.06	32.2	3592	3594	96.5	96.7	0.35	0.351
59	0.2703	0.328	94.97	95	3592	3594	272.3	273.2	0.853	0.853
60	0.2703	0.328	30	30	770.6	770.6	251.8	251.8	0.87	0.87
61	0.2703	0.328	30	30	770.6	770.6	93.6	93.6	0.348	0.348

Comment on Table 1. Pressure agrees to within 0.1% at every one of the 61 state points, and temperature, enthalpy, and entropy each agree to within 1% at about two-thirds of the states and to within 5% at roughly 80% of them, which confirms that the governing equations were reproduced correctly. Two systematic differences remain. First, the calculated mass flow rates run about 13–21% below the published values throughout the AR, AHT, separator, and ORC1 loops (state points 10–25 and 39–61), while the water and ORC2 loop (state points 1–9) matches to within 0.2%; because the paired intensive properties at those same states still agree closely, this points to a difference in the mass basis used to scale the cycle rather than to an error in the balance equations. Second, the largest intensive-property deviations are not scattered randomly but cluster at heat-exchanger outlets – the solution economizer states (22, 24, 25), the humid-air and seawater states of the HDH loops (27–31, 34, 38), and the AHT and preheater outlets (43, 52) – where the enthalpy differences reach roughly 8–26%. This is consistent with the boundary-condition inconsistency described in Section 5: the paper specifies both a UA value and an effectiveness for these units, the two are mutually contradictory, and the effectiveness values adopted here therefore do not reproduce the paper's implied outlet states exactly.

Table 2: State-point comparison — relative humidity, humidity ratio, salinity, and LiBr concentration (calculated vs. base-paper Table 4 values).

State	φ (%)		ω (kg/kg)		Salinity (g/kg)		X_{LiBr} (%)	
	Calc	Paper	Calc	Paper	Calc	Paper	Calc	Paper
1	–	–	–	–	0	0	–	–
2	–	–	–	–	0	0	–	–
3	–	–	–	–	0	0	–	–
4	–	–	–	–	–	–	–	–
5	–	–	–	–	–	–	–	–
6	–	–	–	–	–	–	–	–
7	–	–	–	–	–	–	–	–
8	–	–	–	–	–	–	–	–
9	–	–	–	–	–	–	–	–
10	–	–	–	–	0	0	–	–
11	–	–	–	–	0	0	–	–
12	–	–	–	–	0	0	–	–
13	–	–	–	–	–	–	–	–
14	–	–	–	–	0	0	–	–
15	–	–	–	–	0	0	–	–
16	–	–	–	–	0	0	–	–
17	–	–	–	–	0	0	–	–
18	–	–	–	–	0	0	–	–
19	–	–	–	–	0	0	–	–
20	–	–	–	–	–	–	54.63	54.2
21	–	–	–	–	–	–	54.63	54.2
22	–	–	–	–	–	–	54.63	54.2
23	–	–	–	–	–	–	57.4	56.8
24	–	–	–	–	–	–	57.4	56.8
25	–	–	–	–	–	–	57.4	56.8
26	–	–	–	–	35	35	–	–
27	–	–	–	–	35	35	–	–
28	–	–	–	–	35	35	–	–
29	100	100	0.0256	0.028	–	–	–	–
30	–	–	–	–	35.5	35.6	–	–
31	95	95	0.0594	0.0681	–	–	–	–
32	–	–	–	–	0	0	–	–
33	–	–	–	–	35	35	–	–
34	–	–	–	–	35	35	–	–
35	–	–	–	–	44.9	45.2	–	–
36	56	56	0.028	0.028	–	–	–	–
37	100	100	0.0325	0.0325	–	–	–	–
38	100	100	0.0185	0.0156	–	–	–	–
39	–	–	–	–	–	–	–	–
40	–	–	–	–	–	–	–	–
41	–	–	–	–	–	–	–	–
42	–	–	–	–	–	–	–	–
43	–	–	–	–	–	–	–	–
44	–	–	–	–	0	0	–	–
45	–	–	–	–	–	–	46.95	45.9
46	–	–	–	–	–	–	46.95	45.9
47	–	–	–	–	–	–	48.78	47.8
48	–	–	–	–	–	–	63.38	62.9
49	–	–	–	–	–	–	63.38	62.9
50	–	–	–	–	–	–	63.38	62.9

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Table 2 continued from previous page

State	φ (%)		ω (kg/kg)		Salinity (g/kg)		X_{LiBr} (%)	
	Calc	Paper	Calc	Paper	Calc	Paper	Calc	Paper
51	–	–	–	–	35	35	–	–
52	–	–	–	–	35	35	–	–
53	–	–	–	–	35	100	–	–
54	–	–	–	–	0	0	–	–
55	–	–	–	–	100	100	–	–
56	–	–	–	–	0	0	–	–
57	–	–	–	–	0	0	–	–
58	–	–	–	–	–	–	–	–
59	–	–	–	–	–	–	–	–
60	–	–	–	–	–	–	–	–
61	–	–	–	–	–	–	–	–

Comment on Table 2. The humid-air states (relative humidity, humidity ratio) and the salinity of the seawater/brine states match the published values almost exactly, and the LiBr concentrations although not explicitly stated in the base paper and therefore back-calculated from equilibrium relations in the present model – agree with the published values to within about 1 percentage point at every LiBr state.

4 Results

The following parametric trends were generated from the validated AR and AHT subsystem models and are consistent with the behaviour reported for single-effect LiBr absorption systems in the wider literature [2].

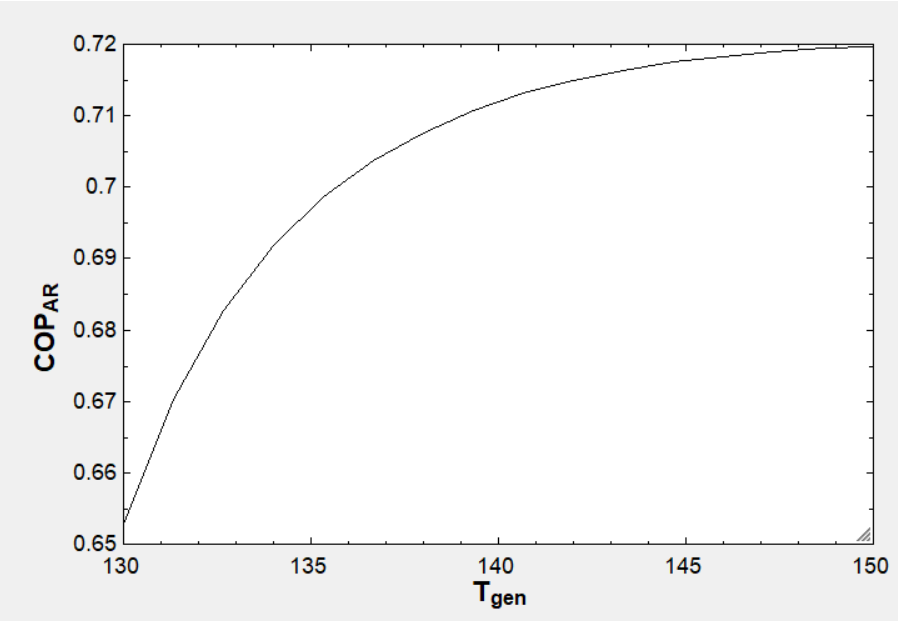


Figure 2: AR coefficient of performance (COP_{AR}) rises and levels off as the generator temperature increases, since a hotter generator drives more refrigerant vapour out of the LiBr solution per unit of heat supplied.

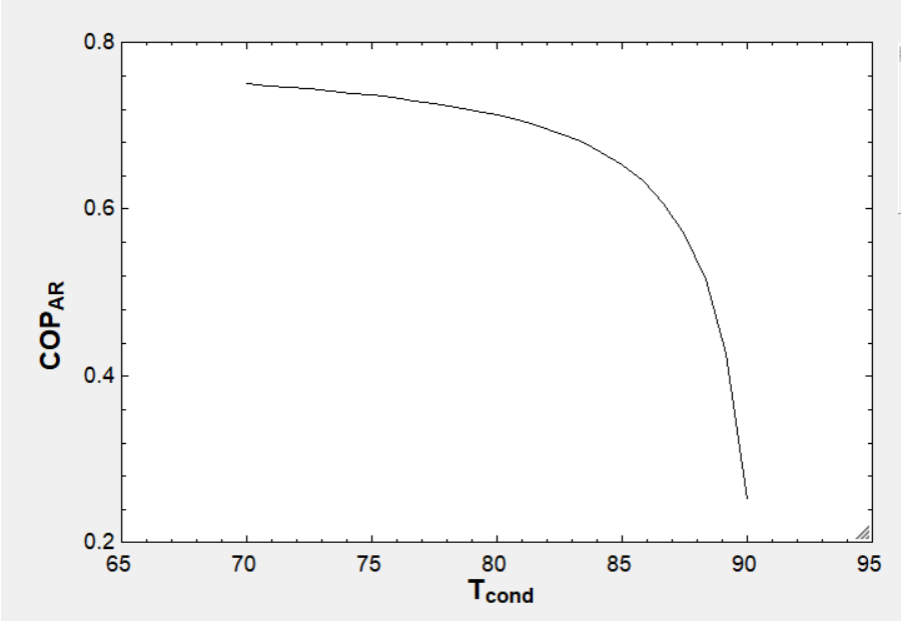


Figure 3: COP_{AR} falls sharply as the condenser temperature increases, because a hotter condenser raises the condensing pressure and therefore the desorption work needed per unit of refrigeration.

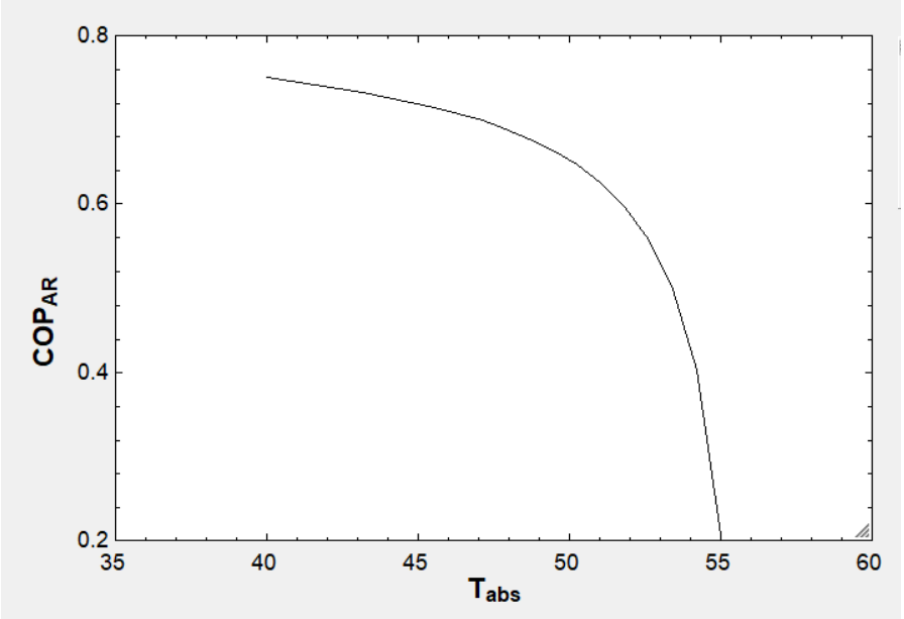


Figure 4: COP_{AR} falls sharply as the absorber temperature increases, since a hotter absorber narrows the strong/weak LiBr concentration difference and reduces refrigerant circulation.

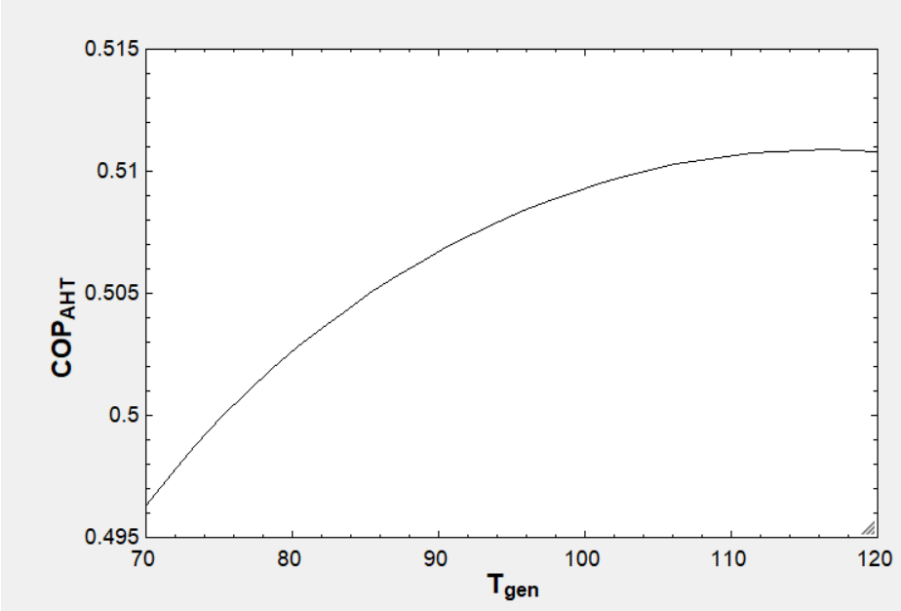


Figure 5: AHT coefficient of performance (COP_{AHT}) increases with generator temperature and then turns over near 115–120°C, reflecting the trade-off between greater heat upgrading and a shrinking absorber temperature lift at high generator temperatures.

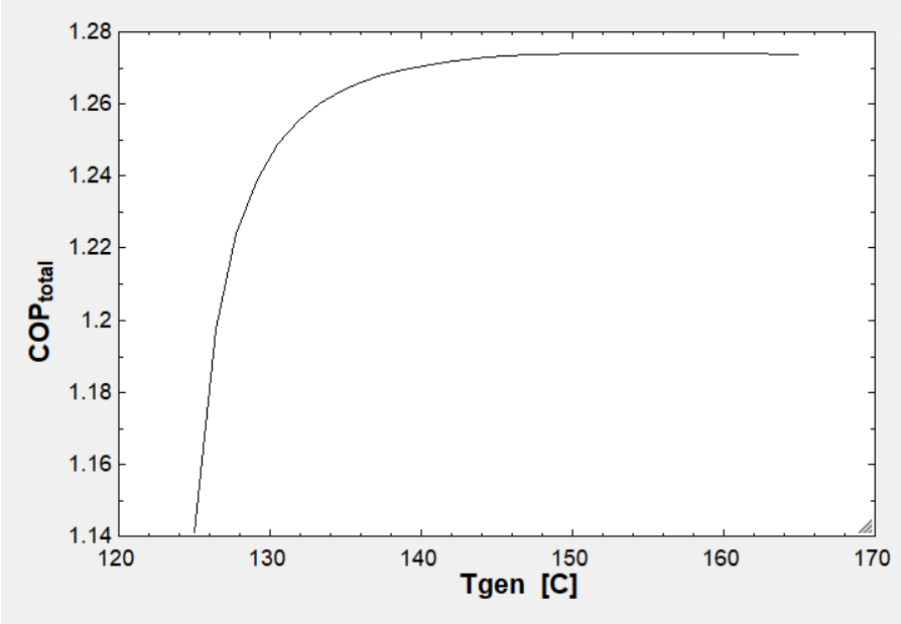


Figure 6: Overall system coefficient of performance (COP_{total}) rises sharply with generator temperature before plateauing, showing that once the AR generator supplies enough heat, performance becomes limited by the downstream AHT/ORC subsystems rather than by the generator itself.

5 Conclusion

The validation process described in Sections 3 and 4 shows close agreement with the base paper's published state-point table and parametric trends overall, but it also surfaced the following inconsistencies and limitations in the base paper itself:

1. The working fluid used for ORC1 is not n-hexane, as stated in the paper; R134a was used instead, since the paper's own reported ORC1 state-point properties (temperature, pressure, enthalpy) are only consistent with R134a and not with n-hexane.
2. The entropy correlation for the built-in LiBr property library (`libr.dll`) used for validation is unavailable in the EES installation used for this study; a literature-based LiBr entropy correlation was implemented directly in the code as a substitute.
3. The paper does not specify the concentration of the strong and weak LiBr solutions; these were back-calculated in the present model from equilibrium relations at each state's temperature and pressure.
4. The paper does not provide sufficient discussion or data to reconstruct most of its parametric studies, so the validation in this report focuses on state-point comparison rather than parametric comparison. Limited parametric checks were nonetheless carried out on the individual subsystems – COP_{AR} vs. T_{cond}, COP_{AR} vs. T_{gen}, COP_{AHT} vs. T_{gen}, and COP_{total} vs. T_{gen} – to confirm that each subsystem follows the trends common to single-effect LiBr absorption systems; these trends align with those reported in the validation study of Modi et al. [2].
5. T_{47} and $X_{s,47}$ are inconsistent guess values for the built-in LiBr property library, so COP_{AHT} vs. T_{REva} and COP_{AHT} vs. T_{Abs} could not be validated for these two variables.
6. As boundary conditions, the paper provides both the overall heat-transfer coefficient (UA) and the effectiveness for its heat exchangers, but these two values are contradictory when checked against the paper's own governing equations. The present model therefore uses the reported effectiveness, rather than UA, as the boundary condition for every heat exchanger.

Despite these issues, the close agreement obtained for nearly every state point in Section 3, and the physically consistent parametric trends in Section 4, support the overall validity of the base paper's reported thermodynamic model, provided the assumptions and corrections listed above are taken into account when the model is reused.

References

[1] S. Salehi and F. Javanfam, "Investigation of a novel solar-assisted multigeneration system comprising water desalination systems, an absorption refrigeration system, a single-effect absorption heat transformer and two organic Rankine cycles," *Heliyon*, vol. 10, p. e36452, 2024, doi: 10.1016/j.heliyon.2024.e36452.

[2] B. Modi, A. Mudgal, and B. Patel, "Energy and exergy investigation of small capacity single effect lithium bromide absorption refrigeration system," *Energy Procedia*, vol. 109, pp. 203–210, 2017, doi: 10.1016/j.egypro.2017.03.040.

[3] M. H. Sharqawy, J. H. Lienhard V, and S. M. Zubair, "Thermophysical properties of seawater: A review of existing correlations and data," *Desalination and Water Treatment*, vol. 16, no. 1–3, pp. 354–380, 2010.